Hyperspectral Imagery Classification Based on Contrastive Learning

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***Abstract*— Supervised machine learning and deep learning methods perform well in hyperspectral image classification. However, hyperspectral images have few labeled samples, which make them difficult to be trained because supervised classification methods rely heavily on sample quantity and quality. Inspired by the idea of self-supervised learning, this article proposes a hyperspectral imagery classification algorithm based on contrast learning, which uses the information of abundant unlabeled samples to alleviate the problem of insufficient label information in hyperspectral data. The algorithm uses a two-stage training strategy. In the first stage, the model is pretrained in the way of self-supervised learning, using a large number of unlabeled samples combined with data enhancement to construct positive and negative sample pairs, and contrastive learning (CL) is carried out. The purpose is to enable the model to make judgments on positive and negative samples. In the second stage, based on the pretrained model, the features of the hyperspectral image are extracted for classification, and a small amount of labeled samples are used to fine-tune the features. Experiments show that the features extracted by self-supervised learning achieved improved results on downstream classification task.**

***Index Terms*— Contrastive learning (CL), hyperspectral imagery classification, self-supervised learning.**

I. Introduction

HYPERSPECTRAL remote sensing technology is based on imaging spectroscopy. It usually uses imaging spectroscopy to acquire remote sensing data, and then processes, analyzes, and applies them [1]–[3]. Hyperspectral imagery (HSI) organically combines the spectral information reflecting the target's radiation properties and the image information reflecting the spatial geometric relationship of the target. Compared with traditional panchromatic imagery and multispectral remote sensing imagery, HSI has a wider spectral range and higher spectral resolution [4], [5]. Its spectrum covers visible light to near-infrared and even mid-infrared regions. Within this range, tens or even hundreds of extremely narrow continuous wavebands are collected, which can finely describe the characters of targets. The subtle differences in the spectral features of ground objects are easier to be distinguished. Therefore, hyperspectral imagery has obvious advantages in target detection and classification.

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Hyperspectral image classification is a technology that assigns a corresponding unique category label to each pixel based on the characteristics of different features [6]. In recent years, many supervised classification methods for hyperspectral images based on machine learning and deep learning have been proposed and achieved good results. Support vector machines (SVMs) have unique advantages in the learning of small samples of high-dimensional data [7]. In [8], the kernel functions of SVMs were combined and adjusted to further improve the classification performance. In addition, K-nearest neighbor (KNN), decision tree (DT), random forests (RFs), convolutional neural networks (CNNs), and other algorithms have been used for the classification of hyperspectral images [9]–[12]. It needs to be noted that spatial context is also very important for remote sensing image processing [13].

The characteristics of hyperspectral images also pose challenge for supervised learning. Under normal circumstances, the classification accuracy will increase with the increase in the dimensionality. However, when the dimensionality exceeds a certain threshold and the number of training samples is insufficient, the accuracy of the parameters estimated by the classifier is often not ideal at this time, so the classification accuracy begins to decline. This phenomenon is called the Hughes phenomenon or the curse of dimensionality [14]. Furthermore, hyperspectral images often have broad scenes and complex surface types, which make them difficult to obtain labels. Classification methods based on supervised learning usually require a large number of labeled samples to train the model, but hyperspectral images often have less data with real sample labels, so it is difficult to accurately estimate the classifier parameters in this case. In response to the above problems, researchers have proposed a series of hyperspectral image classification methods, such as semisupervised learning (SSL), unsupervised learning, transfer learning, and small sample learning for hyperspectral image classification.

SSL trains the model using labeled samples with the assistance of unlabeled samples and makes up for the shortcomings of insufficient labeled samples by mining the information of unlabeled samples. In 1992, the term semi-supervised learning was first proposed and used for classification problems [15]. The semisupervised neural network framework proposed by Ratle *et al.* 16] is a generative model, which can achieve rapid classification of hyperspectral images. However, optimization of generative models is difficult. to solve the dependence of SSL on algorithm parameters, Tuia and Camps-Valls [17] proposed SVM method based on improved linear kernels. The

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training process uses both unlabeled and labeled samples to simplify parameter selection. Ling *et al.* [18] used both labeled and unlabeled samples to construct a loss function to train a semi-supervised convolutional neural network. In addition, the graph-based semi-supervised classification method uses the similarity between samples to spread label information within graph structure, which has good interpretability and excellent classification performance [19].

The unsupervised classification method can be regarded as a clustering process, which divides samples with similar characteristics into the same cluster, and each cluster has a cluster center. Specifically, a certain number of cluster centers are selected first, and each sample is classified into a cluster based on a preset similarity criterion. K-means (K-means), ISODATA, fuzzy C-means (FCM), etc. are the commonly used traditional unsupervised classification algorithms [20]–[22]. In the field of deep learning, stacked autoencoders (SAEs) are trained layer by layer in an unsupervised manner [23]. Data dimension is reduced during the training process, and complex data are expressed by concise high-level features. Finally, the hidden layer output of the encoder is used as the input of the classifier.

Transfer learning refers to the use of knowledge learned from different environment or different field [24], [25]. In machine learning, it can be understood as reusing pretrained models for different tasks. It is worth noting that in deep learning, the pretrained model learns generalized features from source task, and the model needs to be fine-tuned using labeled data in the target task [26], [27]. Zhang *et al.* [28] used two migration strategies to train end-to-end 3-D lightweight convolutional neural networks. One is the cross-sensor, pretraining the model on the source domain hyperspectral dataset with a large number of labeled samples, and the other is the cross-modality, pretraining the model on the 2-D RGB image, and then migrate to the target dataset.

Few-shot learning is the application of meta learning in the field of supervised learning [29], [30]. In the training process, the model learns how to compare sample similarities and how to extract important features and other common parts with different meta-tasks, so the model can also have good results when dealing with new tasks. Based on the prototype network, Tang *et al.* [31] proposed a prototype representation method based on spatial-spectral metric learning to better classify the mixed pixels in hyperspectral images. Liu *et al.* [32] constructed a deep 3-D-CNN model, in which the network is trained by episodes to learn a metric space where samples from the same class are close and those from different classes are far.

If the quantity and category of samples used in SSL are limited, it will limit the generalization performance of the model. The final effect that unsupervised learning methods can achieve is still subject to supervised information. Transfer learning and few-shot learning require a lot of labeled data in the pretraining stage of the pretask. The actual performance of these methods in hyperspectral image classification still needs to be improved. Researchers are expected to use only a small amount of labeled data for training to get a better performance. In this case, self-supervised learning has become a research hotspot. Inspired by self-supervised learning, this article proposes a hyperspectral imagery classification model based on self-supervised contrastive learning (SSCL). The model takes contrastive learning (CL) as the core idea and distinguishes between positive and negative sample pairs as a pretext task in the self-supervised training process. Then it migrates the pretrained model to downstream classification tasks. The contributions of this article are as follows: 1) unsupervised contrastive learning is adopted to learn discriminating features of HSI; 2) conventional spectral-spatial features and deep models are combined to improve the classification accuracy; and 3) experimental results show that CL is an effective strategy to exploit the information of unlabeled pixels.

The rest of this article is organized as follows. Section II introduces self-supervised learning and CL. Section III details the proposed SSCL model. Section IV shows the experimental results and parameter analysis. Section V is a brief conclusion of this article.

II. Related Work

A. Self-Supervised Learning

Self-supervised learning belongs to unsupervised learning, which can be regarded as the "ideal state" of machine learning to a certain extent [33]. Self-supervised learning first constructs appropriate pretext tasks according to data characteristics, using pretext tasks to mine information from a large amount of unlabeled data and automatically generate labels for the data. The network is trained through supervised information constructed in this way, so that it can learn valuable representations for downstream tasks [34]. For downstream tasks, similar to transfer learning, the parameters learned from pretext tasks are transferred for fine-tuning, so as to achieve the ultimate goal. The boundaries between unsupervised learning and self-supervised learning are easy to confuse. From a narrow perspective, unsupervised learning focuses on the detection of specific data patterns, such as anomaly detection, clustering, and dimensionality reduction. The training process does not involve data labels. However, the training process of self-supervised learning involves label information, and its label is automatically generated from pretext tasks. That is to say, self-supervised learning is still essentially in the paradigm of supervised learning [35], [36].

The key to self-supervised learning is how to design pretext tasks to automatically generate labels for data [37]. Pretext tasks can be understood as tasks that are helpful in achieving the final goal. Its advantage is that it simplifies the solution of the target task, aiming at breaking the limitation of manually labeling samples in deep learning. In other words, the purpose of the pretext task is to train the network efficiently without the participation of manual labels. With the generated "pseudo-labels," self-supervised learning implements semantic extraction under the guidance of "pseudo-labels." For example, in computer vision tasks, the commonly used "pseudo-label" generation methods include image rotation, image coloration, image completion, context prediction, etc. [38]–[41].

B. Contrastive Learning

CL is a representative framework in the field of self-supervised learning. The self-supervised learning model based

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on CL constraints feature representation by encoding the similarity of two samples and has made great achievements in computer vision, natural language processing, and other fields [42].

CL first needs to construct positive samples and negative samples. The core idea is to compare the feature space of samples, make the distance between positive sample pairs as close as possible, and enlarge the distance between negative sample pairs. For a positive sample and a certain number of negative samples, the learning purpose of contrast is to make the model distinguish which is positive sample and which are negative samples. CL is essentially representation learning. It will eventually learn a mapping function. For a sample x, it is encoded as $f(x)$. The key to CL is to make $f(x)$ satisfy the following relationship:

$$\text{score}\big(f(x), f\big(x^+\big)\big) \gg \text{score}\big(f(x), f\big(x^-\big)\big). \tag{1}$$

In this formula, x is usually called "anchor" data, x^+ is a positive sample similar to x, x^- is a negative sample not similar to x, and $\text{score}(\cdot,\cdot)$ is a function that measures the similarity of two samples. The similarity measurement function is the basis for designing different contrast loss functions.

To achieve the relationship described by the above formula, a softmax classifier is usually constructed to classify positive and negative sample pairs. This classifier makes $score(\cdot,\cdot)$ tend to assign a larger value to a positive sample pair and a smaller value to a negative sample pair. Its learning goal is to make the feature of x more similar to the feature of x^+ and more dissimilar to the feature of x^-. InfoNCE (Noise Contrastive Estimation) loss function is often used in CL. Suppose we use a dataset $X = [x_1, x_2, \ldots, x_i, \ldots, x_N] \in R^{d\times N}$, where x_i represents the ith training sample, N represents the total number of training data, and d represents the dimension of the data, that is, the number of features. N random samples contain a positive sample and $N-1$ negative samples, then InfoNCE is represented as follows:

$$L_{\text{InfoNCE}} = -E_X\left[\log\frac{\exp\big(s^+\big)}{\exp(s^+) + \sum_{j=1}^{N-1}\exp\big(s_j\big)}\right]. \tag{2}$$

If we use the vector dot product as the similarity measure function, that is,

$$s^+ = \text{score}\big(f(x), f\big(x^+\big)\big) = f(x)^T f\big(x^+\big) \tag{3}$$

$$s_j = \text{score}\big(f(x), f\big(x_j\big)\big) = f(x)^T f\big(x_j\big). \tag{4}$$

Formula (2) is expressed as

$$L_{\text{InfoNCE}} = -E_X \times\left[\log\frac{\exp\big(f(x)^T f\big(x^+\big)\big)}{\exp\big(f(x)^T f(x^+)\big) + \sum_{j=1}^{N-1}\exp\big(f(x)^T f\big(x_j\big)\big)}\right]. \tag{5}$$

InfoNCE was first proposed by Oord *et al.* [43], drawing on the basic idea of noise contrastive estimation (NCE), and used for contrastive predictive coding (CPC). In fact, InfoNCE can be used as an estimate of mutual information, which is the origin of InfoNCE's name. The mutual information of $f(x)$ and $f(x^+)$ indicates the closeness of $f(x)$ and $f(x^+)$.

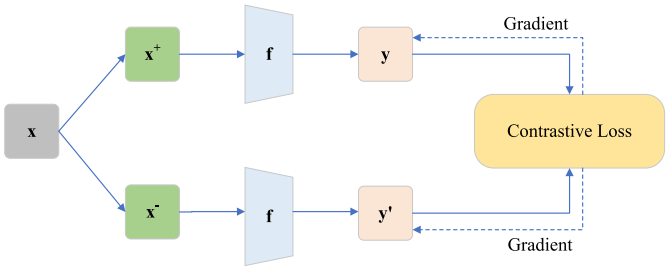


Fig. 1. General framework for CL. For a sample x, x^+ is its similar positive sample and x^- is its nonsimilar negative sample; the contrastive loss makes the distance between positive sample pairs as close as possible and enlarges the distance between negative sample pairs.

Minimizing InfoNCE means maximizing the lower bound of mutual information of $f(x)$ and $f(x^+)$, so that the feature representations of positive sample pairs x^+ and x are more similar. Fig. 1 illustrates a general framework for CL.

III. Proposed SSCL Model

In this section, the structure of the SSCL will be introduced in detail. We will elaborate the proposed hyperspectral image classification algorithm from data processing and model construction. Fig. 2 shows the flowchart of the proposed method.

A. Data Preprocessing

To reduce the computation burden, we first reduce the dimensionality of the original hyperspectral image with principal component analysis (PCA). Then, to make full use of the spatial characteristics of hyperspectral images, the extended morphological profile (EMP) is used to exploit the spatial correlation information of hyperspectral data.

In this article, the original hyperspectral image data are reduced to four dimensions by principal component analysis. PCA is the most adopted method of dimensionality reduction for hyperspectral data. It has the advantages of compressing data volume, reducing computational overhead, removing noise, and having no parameter limitations [44]. Then the morphological profile features are extracted from the principal component image. The basic operations of EMP mainly include corrosion and expansion, opening operation and closing operation, etc. Mathematical morphology analyzes the spatial relationship of pixels with a structural element and can extract image components that are meaningful for expressing and depicting the shape of the region from the image, so that subsequent recognition or classification tasks can capture the most essential features of the target object. In other words, morphological methods can extract the most distinguishable shape features, such as boundaries and connected regions.

The formula for corrosion calculation is as follows:

$$\gamma_B(I) = \delta_B \circ \varepsilon_B(I). \tag{6}$$

The expansion calculation is as follows:

$$\varphi_B(I) = \varepsilon_B \circ \delta_B(I). \tag{7}$$

The opening operation is as follows:

$$\varepsilon_B(I(x)) = \min_{x_i}(I(x_i) \in B_x). \tag{8}$$

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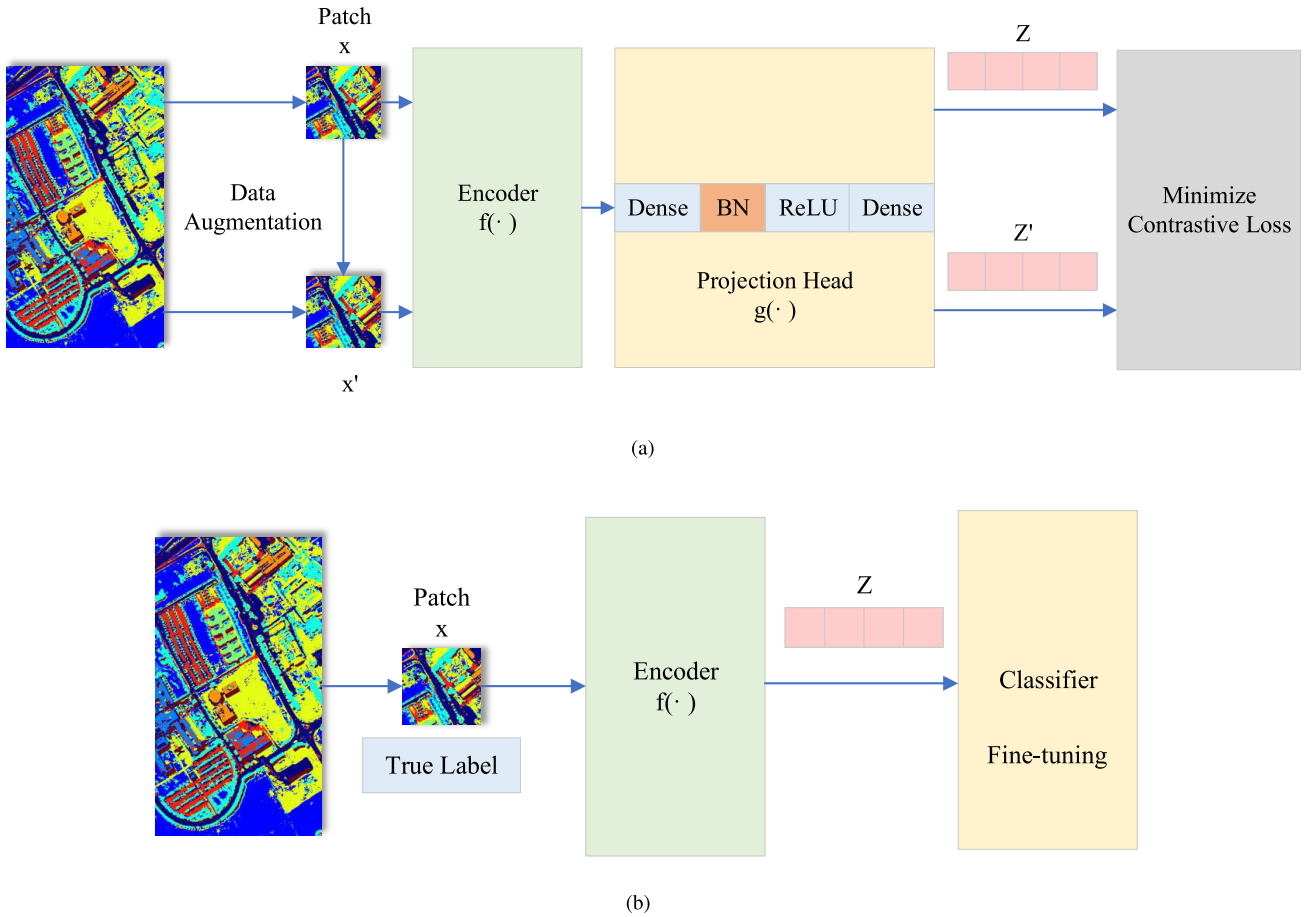


Fig. 2. Flowchart of the SSCL. (a) Pretraining. (b) Fine-tuning.

The closing operation is as follows:

$$\delta_B(I(x)) = \max_{x_i}(I(x_i) \in B_x) \tag{9}$$

where I is the input data, B is the structural element, and B_x is the central value of B at x. The concept of morphological profile is described as follows:

$$\mathrm{MP}^{(n)}(I) = \left[\varphi_r^{(n)}(I), \ldots, \varphi_r^{(1)}(I), I, \gamma_r^{(1)}(I), \ldots, \gamma_r^{(n)}(I)\right] \tag{10}$$

where n represents the number of opening and closing operations performed, I represents the original image to be processed, φ represents the closed operation, and γ represents the open operation.

The formula for extracting morphological profile features from the reduced principal components is as follows:

$$\mathrm{EMP}_k^{(n)}(I) = \left[MP_1^{(n)}(I), \ldots, MP_k^{(n)}(I)\right] \tag{11}$$

where k is the number of principal components after dimensionality reduction.

For a single principal component image, for structural elements of a fixed size, open and close operations get two reconstructed images, the open operation map and the closed operation map. We select four sizes, and each principal component gets eight feature maps; after extracting the first four principal components, a total of $36 = 4 \times (2 \times 4 + 1)$ feature maps will be obtained. The extended morphological features include both spectral and spatial features, which can overcome the limitations of spectral feature and improve the classification performance [45], [46]. In addition, to maintain a consistent data format with the input of the deep convolutional neural network, the processed hyperspectral image data are divided into image blocks of a certain size with each pixel as the center, and the image block contains the spectral and spatial information. This operation introduces the information of the neighboring pixel.

B. Data Augmentation

In this article, data augmentation is implemented by adding random Gaussian noise, and in this way, positive and negative sample pairs participating in training are constructed. Gaussian noise with zero mean can distort the original high-frequency features to a certain extent, and adding appropriate noise can enhance the learning ability of the neural network. The way to produce augmented data is

$$X' = X + \beta \cdot N \tag{12}$$

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where X is the data to be enhanced, N is the random Gaussian noise that obeys the standard normal distribution, and β is set to 1/25.

C. Structure of the SSCL

This section will introduce the proposed algorithm in detail. For the convenience of presentation, assume that the data input into the network in the experiment is represented as $X = [x_1, x_2, \ldots, x_i, \ldots, x_N] \in R^{d\times N}$, where the feature vector of the ith sample is denoted as x_i. The whole algorithm is divided into two stages: pretraining and fine-tuning. In the first stage, the model is pretrained by the CL method, and then the model parameters are saved. In the second stage of the classification task, the model structure is slightly changed and then the supervised learning method is used for fine-tuning to obtain the final classification result.

In the first stage, as shown in Fig. 2, we first perform data augmentation on the dataset X to obtain another set of enhanced data X' with a size of N, and input X and X' into the encoder f to obtain features $f(X)$ and $f(X')$, and then through a series of nonlinear mapping layers, the projected features $g(f(X))$ and $g(f(X'))$ are obtained. The goal of model optimization is to minimize the loss function, so that pairs of positive samples are more similar, and pairs of negative samples are less similar. For the "anchor point" data x_i, we use data augmentation to generate the corresponding positive sample x_i^+. At the same time, the remaining $2N-2$ data except x_i^+ and x_i constitute negative sample pairs with x_i.

To better learn semantic information from the deep network, the algorithm adopts the deep residual network (ResNet50) as the encoder f [47]. The residual network solves the problem: deep networks are usually difficult to train. With the deepening of the network structure, the gradient will gradually decrease or even disappear during the propagation process. For example, using the Sigmoid function as the activation function, each additional layer will attenuate the gradient to 0.25 times of the previous layer during transmission. When the cumulative number of layers increases, the phenomenon of "gradient disappearance" will appear, and the previous network layer parameters cannot be effectively trained. The residual network is obtained by cascading the basic residual units, and the output of the residual unit is obtained from the sum of the output and input of multiple convolutional layers cascaded and then subjected to ReLU nonlinear activation. For a deeper network, the dimension close to the output layer of the network is large, resulting in a huge amount of calculation. The structure of Fig. 3 is adopted, and the dimension is reduced by 1×1 convolution first, and then 3×3 convolution later, a 1×1 convolution is used to increase the dimension and restore to the original dimension. This structure greatly reduces the amount of calculation. For the input data x, assuming that the hidden layer is expressed as $H(x)$, according to the structure of the residual unit, $H(x)$ is expressed as the sum of the forward neural network $F(x)$ and x, that is, $H(x) = F(x) + x$.

After the encoder, the structure of FC - BN - ReLU - FC is used as the projection head, and the output $f(X)$ obtained by the encoder is mapped to $g(f(X))$. The function is to map

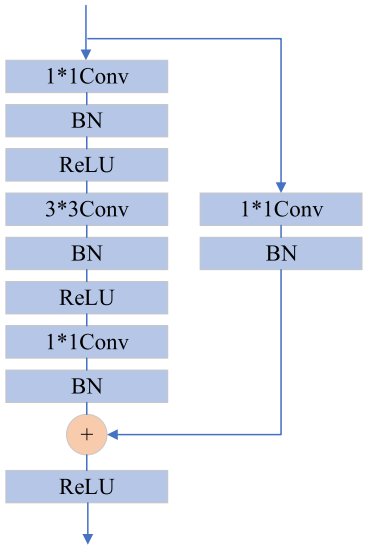


Fig. 3. Structure of the ResBlock.

the representation obtained by the encoder to the feature space applying contrastive loss, helping to identify the invariant features of each input. The calculation formula of the projection head is

$$g(f(X)) = W^{(2)}\left(\sigma\left(W^{(1)}(f(X))\right)\right) \tag{13}$$

where $W^{(1)}$ and $W^{(2)}$ are the weight parameters of the two fully connected layers respectively, and σ is the ReLU activation function.

The contrastive loss adopts the form of NCE loss function, and the similarity measurement adopts cosine similarity, which is specifically expressed as

$$L_{i,j} = -\log\frac{\exp\left(\text{score}\left(z_i, z_j\right)/\tau\right)}{\sum_{k=1}^{2N}\mathbb{I}_{[k\neq i]}\exp(\text{score}(z_i, z_k)/\tau)} \tag{14}$$

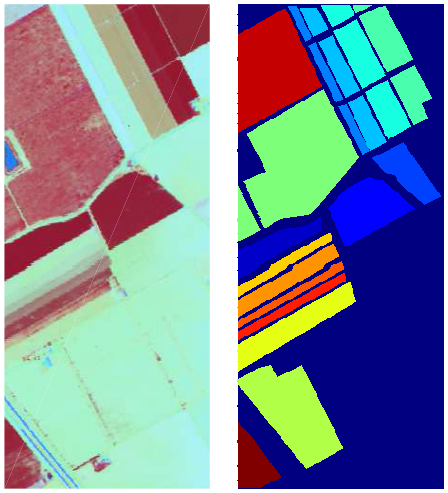
where z_i and z_j are the projection output of (13) of the input features x_i and x_j. For convenience, $(2--1.2m)$ are used as the positive pair index, and then a final symmetrized loss function can be defined as follows:

$$L = \frac{1}{2N}\sum_{m=1}^{N}\left[L_{2m-1,2m} + L_{2m,2m-1}\right]. \tag{15}$$

Among them, τ is an adjustable temperature parameter; $\mathbb{I}_{[k\neq i]}$ is an indicator function, and the value is 0 if and only if $k = i$; otherwise, it is 1; the denominator contains $2N-1$ terms, of which the jth sample is a positive sample, and the other $2N-2$ items are negative samples. The loss function is based on l_2 regularization, using the *cosine* function to express regularization: $\text{cosine}(z_i, z_j) = z_i^T z_j/\|z_i\|\|z_j\|$. Due to the introduction of the temperature parameter τ, this loss function is also called normalized temperature-scale cross-entropy (NT-Xent) loss [48].

The purpose of minimizing the loss function is to make the similarity of the positive sample pairs z_i and z_j as high

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(a) (b)

Fig. 4. Three-band color composing image and ground truth of Salinas. (a) Three-band color composite image. (b) Ground truth.

as possible. In the classification stage, the encoder parameters in the pretrained network are fixed. The projection head is discarded, and a fully connected layer is added as a classifier for supervised training.

IV. EXPERIMENTAL RESULTS

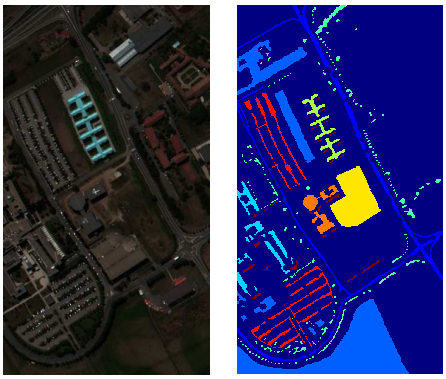
A. Experimental Setup

1) Hyperspectral Datasets: To demonstrate the effectiveness of the proposed SSCL method, we will test the proposed method on three widely used hyperspectral datasets: Salinas image, Pavia University image, and Botswana image.

The Salinas dataset contains 512×217 samples. It contains 224 bands, and the wavelength range is 400–2500 nm. The image has the characteristics of high-altitude resolution, with a spatial resolution of 3.7 m. The 108–112, 154–167, and 224 bands are absorbed by water. To ensure the accuracy of the results, these bands are usually discarded in the experiment, and only the remaining bands are used for research. The Salinas dataset mainly contains 16 classes. Fig. 4 shows the pseudo-color image and ground-truth image of the Salinas dataset.

The Pavia University image was acquired by ROSIS-03, an imaging spectrometer based on a reflective optical system. The number of spectral bands is 115, the wavelength range is 430–860 nm, the size is 610×340 pixels, and the spatial resolution is 1.3 m. After discarding the most noisy frequency bands, 103 bands will be used for research. There are a total of 42 776 samples in the dataset, covering nine feature types. The pseudo-color map and ground truth of this dataset are shown in Fig. 5.

The Botswana dataset was acquired in the Okavango Delta of Botswana. Its size is 620×340, the number of bands is 242, and the spatial resolution is 30 m. After removing the water absorption band and the high-noise band, there are only 145 bands left in the candidate feature set. The data used in the experiment were obtained in May 2001 and contain a total of 14 types of ground classes, covering seasonal swamps,



(a) (b)

Fig. 5. Three-band color composing image and ground truth of Pavia University. (a) Three-band color composite image. (b) Ground truth.



(a) (b)

Fig. 6. Three-band color composing image and ground truth of Botswana. (a) Three-band color composite image. (b) Ground truth.

occasional swamps, and dry woodlands. The pseudo-color map and label information of the Botswana dataset are shown in Fig. 6.

2) Evaluation Indexes: There are many performance indicators used to evaluate the classifier in hyperspectral image classification. Three most commonly used indicators are adopted in this article: overall accuracy (OA), average accuracy (AA), and Kappa coefficient. OA refers to the percentage of all correctly classified samples in all the test samples. AA refers to the average of the classification accuracy of each category. The Kappa coefficient is used in statistics to measure consistency. For classification problems, consistency refers to the consistency between the predicted results of the model and the actual classification results. The Kappa coefficient can indicate the degree of error reduction of the classification result compared with the completely random classification. In practice, the range is usually $[0, 1]$. The larger the value, the better the classification performance of the model.

B. Classification Accuracy

In the experiment, 30% samples of each category are randomly selected for the first stage of CL pretraining. For the second stage of supervised learning, 10% samples of each category are randomly selected as the training set, and the remaining samples are used to test. The number of training iterations is 100 rounds, and the parameter τ in the loss function is set to 0.5. For the Salinas and Pavia University datasets, the input image block size is set to 23×23, while the Botswana dataset is set to 5×5. In the classification stage, 10% samples of the original dataset are randomly selected to fine-tune the model, and other data are used to test. Table I shows the number of samples contained in the training set and the test set.

To evaluate the performance of the proposed method more intuitively, six common hyperspectral image classification

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TABLE I
NUMBER OF SAMPLES FOR TRAINING AND TEST OF EACH CATEGORY

Salinas				Pavia University				Botswana			
No.	Categories	Train	Test	No.	Categories	Train	Test	No.	Categories	Train	Test
C1	Brocoli green weeds 1	201	1808	C1	Asphalt	663	5968	C1	Water	27	243
C2	Brocoli green weeds 2	373	3353	C2	Meadows	1865	16784	C2	Hippo grass	10	91
C3	Fallow	197	1779	C3	Gravel	210	1889	C3	Floodplain grasses1	25	226
C4	Fallow rough plow	139	1255	C4	Trees	306	2758	C4	Floodplain grasses2 marsh	22	193
C5	Fallow smooth	268	2410	C5	Metal sheets	135	1210	C5	Reeds1	27	242
C6	Stubble	396	3563	C6	Bare soil	503	4526	C6	Riparian	12	217
C7	Celery	358	3221	C7	Bitumen	133	1197	C7	Firecar	26	233
C8	Grapes untrained	1127	10144	C8	Bricks	368	3314	C8	Island interior	20	183
C9	Soil vinyard develop	620	5583	C9	Shadows	95	852	C9	Acacia woodlands	31	283
C10	Corn senesced green weeds	328	2950					C10	Acacia shrublands	25	223
C11	Lettuce romaine 4wk	107	961					C11	Acacia grasslands	31	274
C12	Lettuce romaine 5wk	193	1734					C12	Short mopane	18	163
C13	Lettuce romaine 6wk	91	825					C13	Mixed mopane	27	241
C14	Lettuce romaine 7wk	107	963					C14	Exposed soils	9	86
C15	Vinyard untrained	727	6541								
C16	Vinyard vertical trellis	181	1626								
Total		5413	48716	Total		4278	38498	Total		325	2923

TABLE II
CLASSIFICATION RESULTS(%) OF THE SALINAS DATASET

Categories	PCA-EPFs	EMP-SVM	2D-CNNs	MLP-CNNs	ResNet50	SANI-SSL	SSCL
Brocoli green weeds 1	**100.00**	98.31	99.04	**100.00**	**100.00**	**100.00**	**100.00**
Brocoli green weeds 2	98.37	97.25	98.76	99.22	98.34	**100.00**	99.97
Fallow	99.18	97.38	**100.00**	98.34	97.63	**100.00**	**100.00**
Fallow rough plow	98.52	98.23	97.53	97.36	98.26	99.52	**99.92**
Fallow smooth	97.43	97.17	97.22	98.19	98.61	99.25	**99.96**
Stubble	98.67	98.51	97.60	98.51	**100.00**	**100.00**	**100.00**
Celery	97.93	97.59	98.43	97.55	97.86	**99.84**	99.32
Grapes untrained	93.72	97.73	97.18	97.72	98.03	94.45	**99.48**
Soil vinyard develop	97.62	96.72	97.45	98.16	98.45	**100.00**	**100.00**
Corn senesced green weeds	98.45	96.84	97.96	98.31	97.54	99.22	**99.66**
Lettuce romaine 4wk	98.28	97.10	97.84	**100.00**	98.40	99.06	96.77
Lettuce romaine 5wk	98.71	97.39	98.95	98.46	**100.00**	**100.00**	**100.00**
Lettuce romaine 6wk	97.66	97.35	97.42	98.18	98.31	97.82	**100.00**
Lettuce romaine 7wk	98.74	**100.00**	98.35	98.37	99.28	98.34	**100.00**
Vinyard untrained	92.33	95.16	93.23	94.05	97.73	95.49	**99.62**
Vinyard vertical trellis	98.10	96.77	97.51	98.23	98.52	**100.00**	**100.00**
OA	98.46	97.54	98.19	98.21	98.32	98.04	**99.70**
AA	98.13	97.34	97.78	98.17	98.55	98.94	**99.67**
Kappa	0.9785	0.9742	0.9813	0.9822	0.9814	0.9782	**0.9967**

algorithms are used for comparison, namely, PCA-EPFs [49], EMP-SVM [45], 2-D-CNNs [50], MLP-CNNs [51], ResNet50 [52], and SANI-SSL [53]. All the experiments were performed on the 3.2-GHz CPU, GeForce GTX 1060 GPU, and 16-GB memory computer with Pytorch.

Table II shows the experimental results of the Salinas dataset. It can be seen that the proposed SSCL method shows the best results in terms of OA, AA, and Kappa coefficient. The proposed method and ResNet50 method use ResNet50 as the feature extractor, and OA is improved by nearly 1.4% on the basis of it. The two deep neural network structures, 2D-CNNs and MLP-CNNs, are also used for comparison. The classification results in some categories are not as good as the traditional classification methods PCA-EPFs, EMP-SVM, and semi-supervised method SANI-SSL. This situation may be caused by the imbalance between the classes of the dataset. This is also one of the possible reasons why PCA-EPFs outperforms the proposed method in some categories. PCA-EPFs make the image smoother through edge-preserving filtering and obtain the best results except SSCL. Fig. 7 visually demonstrates the classification performance of different algorithms.

Table III lists the classification results of the Pavia University dataset. It can be seen that the proposed method shows the best performance in all the three indicators. In terms of OA, compared with the EMP-SVM method that achieved the worst

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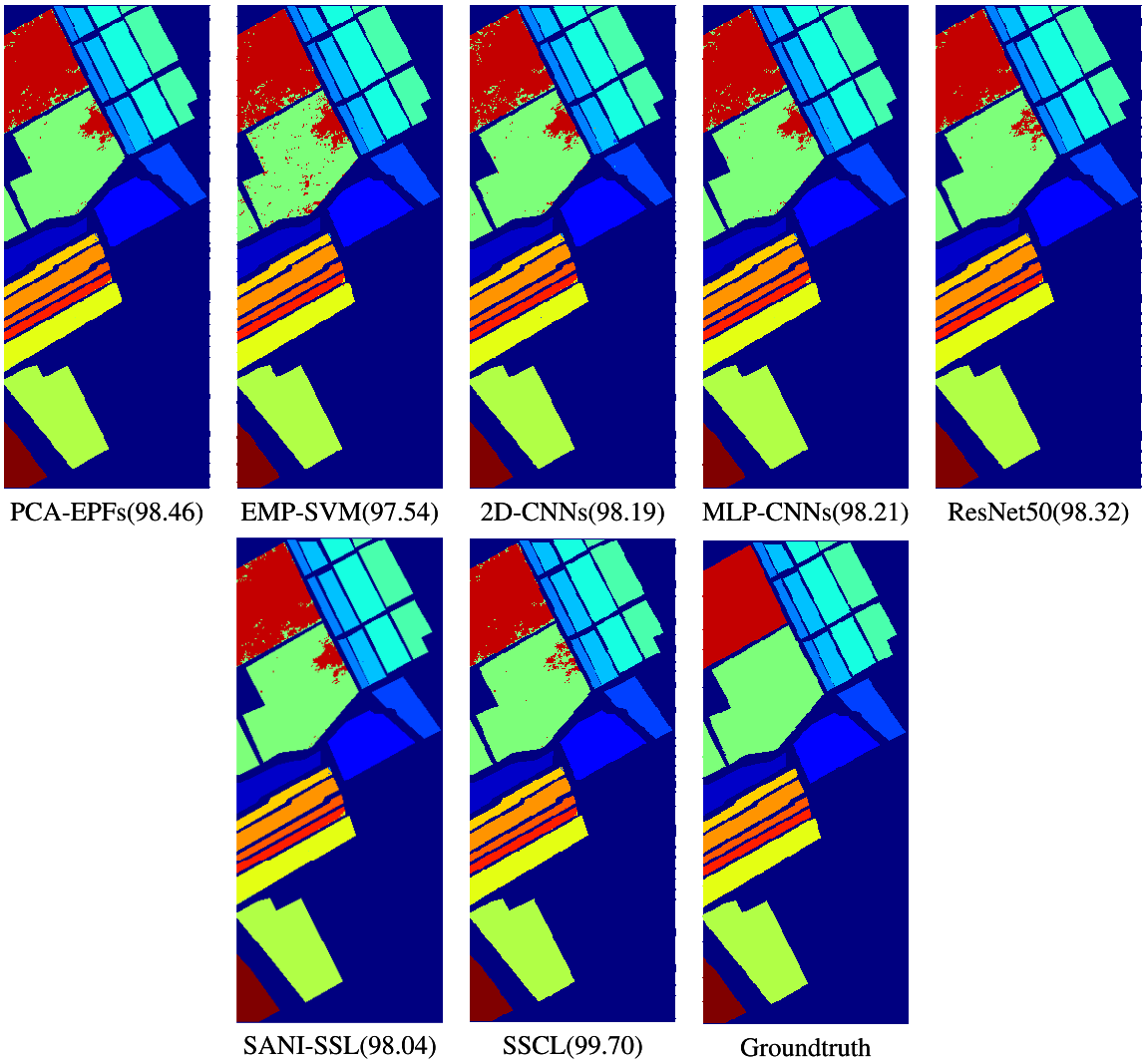


Fig. 7. Classification maps of all the methods on the Salinas dataset.

TABLE III
CLASSIFICATION RESULTS(%) OF THE PAVIA UNIVERSITY DATASET

Categories	PCA-EPFs	EMP-SVM	2D-CNNs	MLP-CNNs	ResNet50	SANI-SSL	SSCL
Asphalt	94.85	93.52	95.63	94.97	95.23	97.08	**97.87**
Meadows	**100.00**	95.25	95.24	95.32	**100.00**	98.30	98.68
Gravel	89.18	87.80	88.71	88.89	90.45	**90.74**	89.47
Trees	93.56	94.27	93.74	95.85	**96.19**	94.37	92.42
Metal sheets	**100.00**	96.34	96.45	97.24	96.33	98.92	98.84
Bare soil	94.52	95.92	95.38	96.60	95.71	95.92	**97.75**
Bitumen	94.37	95.62	**96.42**	95.23	93.92	95.25	96.07
Bricks	92.43	94.39	94.73	95.36	**96.31**	95.49	96.11
Shadows	95.81	96.56	96.67	96.71	96.28	96.04	**96.83**
OA	96.02	95.34	96.29	95.93	96.14	96.26	**97.21**
AA	94.97	94.41	94.77	95.13	95.60	95.79	**96.00**
Kappa	0.9587	0.9494	0.9570	0.9583	0.9611	0.9513	**0.9630**

results, the proposed method improved by 1.87%. 2-D-CNNs achieved the highest OA other than SSCL. In addition, considering the three indicators, the overall performance of ResNet50 and SANI-SSL exceeds other methods except SSCL. The above experimental results show that the self-supervised model based on CL can extract discriminative features that are more conducive to classification. The classification maps are provided in Fig. 8.

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TABLE IV
CLASSIFICATION RESULTS(%) OF THE BOTSWANA DATASET

Categories	PCA-EPFs	EMP-SVM	2D-CNNs	MLP-CNNs	ResNet50	SANI-SSL	SSCL
Water	**100.00**	99.10	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
Hippo grass	**100.00**	91.21	98.80	98.90	99.21	97.80	**100.00**
Floodplain grasses1	**100.00**	**100.00**	**100.00**	99.37	98.05	**100.00**	98.67
Floodplain grasses2 marsh	**97.42**	95.88	95.36	96.94	96.06	**97.42**	95.85
Reeds1	89.95	87.60	90.91	**94.86**	91.18	91.32	91.74
Riparian	93.80	89.26	91.91	94.21	94.32	90.91	**95.87**
Firecar	99.42	99.57	95.57	97.35	99.29	99.57	**100.00**
Island interior	**100.00**	**100.00**	**100.00**	97.98	98.97	**100.00**	**100.00**
Acacia woodlands	95.76	92.75	95.76	95.17	95.76	**96.47**	95.05
Acacia shrublands	99.10	97.31	**100.00**	97.38	98.94	95.07	**100.00**
Acacia grasslands	95.96	96.35	94.61	95.52	97.73	97.81	**98.54**
Short mopane	**100.00**	99.39	99.39	97.80	98.94	99.39	**100.00**
Mixed mopane	98.34	97.10	96.68	97.93	99.18	97.51	**100.00**
Exposed soils	97.20	100.00	99.71	96.41	96.20	**100.00**	96.51
OA	97.06	95.69	96.61	96.77	97.09	97.10	**97.96**
AA	97.68	96.11	97.05	97.13	97.42	97.38	**98.02**
Kappa	0.9704	0.9533	0.9633	0.9751	0.9692	0.9685	**0.9770**

TABLE V
TIME CONSUMPTION OF DIFFERENT METHODS WITH SALINAS SCENE (SECONDS)

PCA-EPFs	EMP-SVM	2D-CNNs	MLP-CNNs	ResNet50	SANI-SSL	SSCL
9.6	22.4	2637.3	2385.6	3083.8	43.9	7408.9

TABLE VI
OA (%) OF DIFFERENT τ VALUES FOR THE SALINAS, PAVIA UNIVERSITY, AND BOTSWANA DATASETS

	0.01	0.05	0.1	0.5	0.8	1.0
Salinas	96.58	97.32	98.49	99.70	98.83	97.33
Pavia University	95.34	95.75	96.20	97.21	96.62	95.37
Botswana	95.06	95.89	96.02	97.88	97.17	95.51

Table IV lists the experimental results on the Botswana dataset. Compared with other comparison algorithms, the proposed SSCL still shows the best results with an accuracy of 97.96%. The basic network structure of 2-D-CNNs and MLP-CNNs is 2-D convolutional neural networks, and the average accuracy of the two methods is close, 96.61% and 96.77%, respectively. The classification maps are also provided in Fig. 9 for visual comparison.

Besides the classification accuracy, the computation burden is also important for HSI classification methods, so the consuming time of all the methods with Salinas scene is listed in Table V. We can find that the deep-learning-based methods consume much longer time than conventional methods. For the proposed SSCL method, because of the introduction of self-supervised learning with abundant unlabeled samples, it consumes the longest time.

C. Parameters Analysis

1) Effect of Temperature Parameter: An adjustable temperature parameter τ is included in the loss function, with a range of $[-1, 1]$, whose function is to scale the input and expand the range of cosine similarity. Appropriate adjustment of the temperature parameter τ will affect the experimental results. Table VI shows OA on the three datasets when the parameter τ takes different values. In the experiment, 10% of each category of data is randomly selected as the labeled training set. The neighborhood size of the Salinas and Botswana datasets is 23×23, and that of the Botswana dataset is 5×5. The experimental results show that the best classification result is obtained when the value of τ is 0.5.

2) Effect of the Neighborhood Size: Taking each pixel as the center to take a certain size of image block instead of individual pixels as training data, this operation introduces spatial information. For hyperspectral data, adjacent pixels have a high probability of belonging to the same category or showing similar characteristics. Choosing different neighborhood sizes will affect the experimental results to a certain extent. Table VII lists OA of the proposed model with different neighborhood sizes. In the experiment, the parameter τ is set to 0.5, and 10% samples of the labeled data are randomly selected as the training set. It can be seen from Table VII that for the Salinas and Pavia University datasets,

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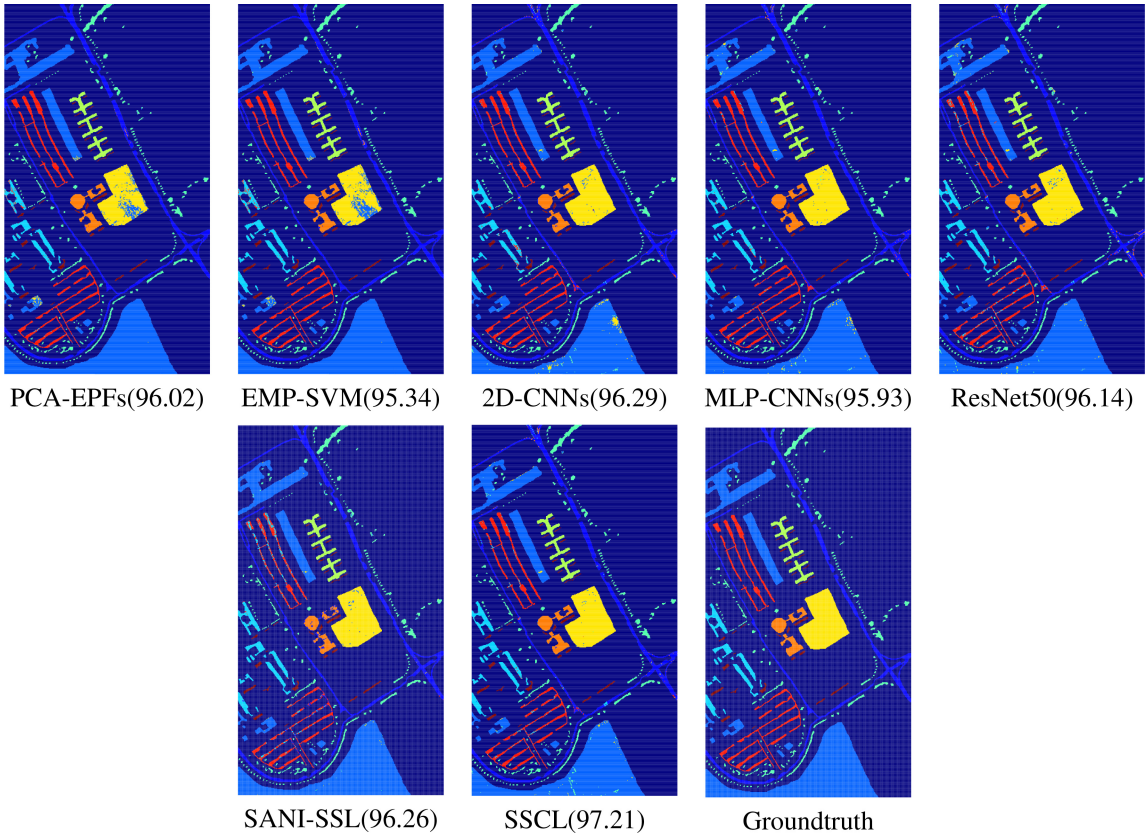


Fig. 8. Classification maps of all the methods on the Pavia University Dataset.

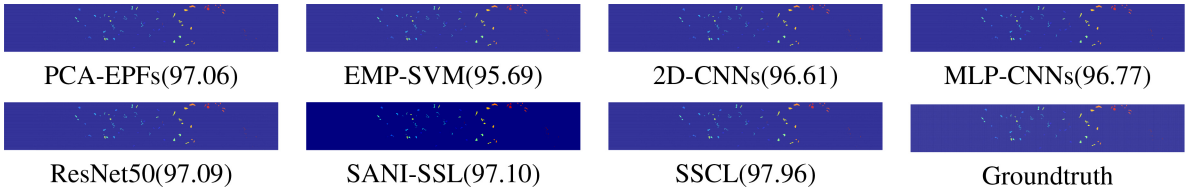


Fig. 9. Classification maps of all methods on the Botswana dataset.

TABLE VII
OA (%) OF DIFFERENT NEIGHBORHOOD SIZES FOR THE SALINAS, PAVIA UNIVERSITY, AND BOTSWANA DATASETS

	3×3	5×5	7×7	11×11	23×23
Salinas	95.86	97.13	97.75	98.34	99.70
Pavia University	95.02	95.74	96.28	96.53	97.21
Botswana	93.60	97.88	94.87	91.82	89.70

as the neighborhood size increases, the OA increases, but for Botswana, when the neighborhood size exceeds 5×5, the OA decreases as the neighborhood increases. Observing the three datasets, it can be found that Botswana is significantly different from the other two datasets. Its covered area is larger and the sample distribution is sparse. Therefore, it is reasonable to speculate that a large number of background pixels are introduced when taking a larger neighborhood size for the Botswana dataset. However, the training process involves data enhancement of neighboring blocks, and if background pixels become the dominant factor, which adds too much noise to the data, resulting in a decline in the classification result. According to the results of Table VII, the appropriate neighborhood size in the experiment is set to 23×23 for the Salinas and Pavia University datasets, and 5×5 for the Botswana dataset.

D. Performance of SSCL Under Different Training Ratios

To verify the classification effect of the proposed method in the case of fewer labeled samples, different training sample ratios are set in the fine-tuning stage to conduct experiments on four deep neural network models. Figs. 10–12 shows the OA of the 2-D-CNNs, MLP-CNNs, ResNet50, and SSCL on three datasets when different training ratios of samples are taken.

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TABLE VIII
OA OF SALINAS SCENE WITH DIFFERENT BATCH SIZES

Batchsize	8	16	32	64	128	256
Overall accuracy	94.87	96.92	98.75	99.29	99.70	99.51

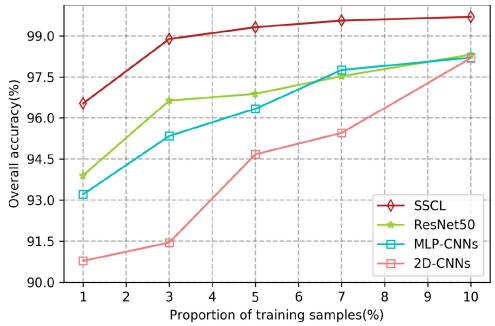


Fig. 10. OA of three contrast algorithms and SSCL for Salinas dataset at different training ratios.

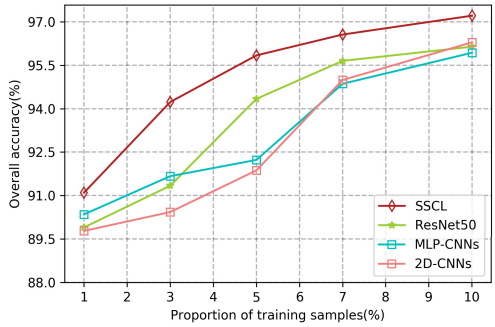


Fig. 11. OA of three contrast algorithms and SSCL for Pavia University dataset at different training ratios.

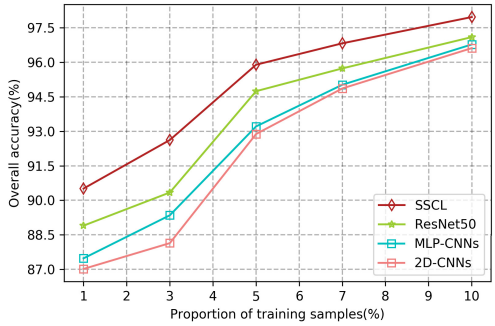


Fig. 12. OA of three contrast algorithms and SSCL for Botswana dataset at different training ratios.

In the experiment, the temperature parameter τ is set to 0.5, the neighborhood size of the Botswana dataset is set to 5×5, and for the remaining two datasets, 23×23. From Fig. 10, it can be seen that when the ratio of the training samples is 1%, the classification result of SSCL for the Salinas dataset is close to the result of ResNet50 training with a ratio of 3% and 5%. For the Pavia University dataset, as shown in Fig. 11, when the training sample is 3%, compared with ResNet50 and 2-D-CNNs, the OA of SSCL improves the most. When the ratio is taken at 5%, the lifting effect of SSCL is more obvious compared with MLP-CNNs. The classification results of the Botswana dataset also reflect the advantages of SSCL when the training samples are 1% and 3%. Based on the above analysis, it can be concluded that SSCL has greater advantages in the case of fewer labeled samples and can achieve higher classification accuracy.

Besides the above-mentioned parameters, the batch size is an important parameter for the CL-based method. We listed the experimental results with Salinas scene in Table VIII. We can see that batch size has an apparent influence in the SSCL method; with the increase in batch size, the classification accuracy promoted effectively. However, when the batch size exceeds some threshold value, the accuracy has a small decrease. So, the batch size is set as 128 in our experiment.

V. CONCLUSION

In this article, SSCL is proposed for hyperspectral imagery classification, which is a deep learning model based on self-supervised learning. The proposed algorithm is a two-stage model. In the pretraining stage, data augmentation is performed first, and then positive and negative sample pairs are constructed using the original and generated data. In this model, CL is a pretext task of self-supervision. The core idea is to make the model have the ability to distinguish between positive and negative sample pairs. The ultimate goal of the pretraining process is to minimize the distance between positive sample pairs and make the distance between negative sample pairs as large as possible. In the second stage, hyperspectral image classification is used as the downstream task, the parameters of the pretrained model are retained, and the encoder is used for feature extraction. Experiments proved that the hyperspectral image classification framework based on CL is effective and has advantages when the number of labeled samples is small, indicating that the proposed model can learn effective feature representation by CL.

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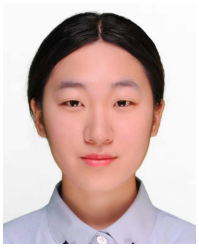
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